Cost-Constrained Edge Removal for Epidemic Containment on Social Networks: A Comparison of Heuristic and GNN-Learned Policies on Facebook100

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Abstract

We study cost-constrained epidemic containment on real social networks (Facebook100) under a stochastic Susceptible-Infected-Susceptible (SIS) model. Edge removal is the intervention modality and the cost of removing an edge is derived from the similarity of the two endpoints' feature vectors via a sigmoid-dot-product proxy. We compare four policies: random, edge betweenness over cost, a Louvain-community "distance threshold" (the realistic implementable form), and a graph neural network (GNN) policy that learns node embeddings whose similarity is the per-edge cost. We sweep cost budgets to recover Pareto frontiers of cost vs. steady-state prevalence on five Facebook100 campuses, run robustness panels (SIR alternative, imperfect compliance, adversarial seeding) and three ablations (GNN architecture, cost function, synthetic graph topology), and fit a cross-campus regression of policy AUC against descriptive graph statistics. We discuss when the GNN-learned cost beats the hand-designed cost, when classical betweenness suffices, and how much we lose by constraining ourselves to enforceable rules.

1 Introduction

Scientific question. *Given a social contact graph in which vertices carry feature information and edges carry costs derived from endpoint feature similarity, what is the minimum-cost set of edges whose removal keeps the steady-state SIS prevalence below a fixed threshold?*

Motivation for the cost-aware reframe. The classical literature on epidemic containment via network modification studies the *cardinality* version of this problem: remove the minimum number of edges, or equivalently the maximum-impact k edges, such that the residual graph's spectral radius drops below the epidemic threshold $\tau_c = 1/\lambda_1(A)$ (Wang et al., 2003; Tong et al., 2012; Saha et al., 2015). The cardinality formulation treats every edge as identical. In real social graphs this is unrealistic: cutting the friendship between two roommates is socially much costlier than cutting an acquaintance tie across a campus. A containment policy that ignores this asymmetry is either practically impossible to deploy (it severs critical ties) or underutilizes cheap interventions (it misses harmless cuts that would still bend the curve). The cost-aware reframe takes node features as input, derives a per-edge cost from those features, and minimizes the total cost of the edges removed subject to a containment constraint.

Contributions.

1. An end-to-end empirical comparison of four edge-removal policies (random, edge betweenness over cost, a Louvain-induced community-distance threshold, and a GNN-learned cost policy) on five Facebook100 campuses chosen for diversity in clustering coefficient.

2. A graph neural network policy whose per-edge cost $c_e^{\text{GNN}} = \sigma(\langle z_u, z_v \rangle)$ is *learned* from a self-supervised link-prediction objective on the friendship graph itself. The GNN converts attribute-driven homophily into a continuous cost surface, and the learned cost is combined with NetMelt-style spectral edge importance into a single bang-for-buck score.

3. An implementability analysis: we report the gap between the ideal policy (GNN over cost) and a realistic, enforceable policy (community-distance threshold), under both perfect and imperfect compliance.

4. Three robustness panels (SIR alternative, 80% compliance, adversarial top-eigenvector seeding) and three ablations (GNN architecture, cost function, synthetic graph topology) that characterize *when* each policy wins.

2 Background

2.1 Graph and SIS dynamics

Let $G = (V, E)$ be an undirected graph with $n = |V|$ vertices and $m = |E|$ edges, and let $A \in \{0,1\}^{n \times n}$ be its adjacency matrix. Each vertex $v \in V$ carries a feature vector $x_v \in \mathbb{R}^d$ (in our case the L2-normalized one-hot encoding of seven Facebook100 attributes). At each discrete time step, every infected node infects each susceptible neighbor independently with probability β per edge, and recovers (back to susceptible) with probability γ. Equivalently, a susceptible vertex with k infected neighbors becomes infected with probability $1 - (1-\beta)^k$.

The SIS model has a phase transition governed by the leading eigenvalue of A. Wang et al. (2003) showed that the quenched mean-field epidemic threshold is

$$\tau_c = \frac{1}{\lambda_1(A)}.$$

For $\beta/\gamma < \tau_c$ the epidemic dies out almost surely; for $\beta/\gamma > \tau_c$ a positive-prevalence endemic state persists. Controlling the leading eigenvalue therefore controls containment: removing edges that contribute most to $\lambda_1(A)$ is the classical recipe for raising τ_c above β/γ.

We work in the stochastic (per-vertex Bernoulli) version of SIS and define our primary outcome as the time-averaged *steady-state prevalence*: the fraction of infected vertices averaged over a post-burn-in window of the simulation.

2.2 Cost-aware edge removal as a constrained optimization problem

Let $c : E \to \mathbb{R}_{\geq 0}$ be a per-edge cost function. For any subset $S \subseteq E$ define the total cost $C(S) = \sum_{e \in S} c_e$ and write $G \setminus S$ for the residual graph after removing the edges in S. Let $\text{Outcome}(G_S)$ be the steady-state prevalence of an SIS process on $G \setminus S$ at fixed (β, γ). The constrained optimization problem we study is

$$\min_{S \subseteq E} \quad C(S) \qquad \text{s.t.} \quad \text{Outcome}(G \setminus S) \leq \tau,$$

or equivalently in budgeted form

$$\min_{S \subseteq E} \quad \text{Outcome}(G \setminus S) \qquad \text{s.t.} \quad C(S) \leq B,$$

where τ is a containment target and B a cost budget. We sweep B over a logarithmic grid and report the implied Pareto frontier. This is a generalization of the cardinality formulation of Tong et al. (2012) and Saha et al. (2015); the non-uniform costs are what make the problem genuinely cost-aware.

2.3 The cost proxy

We follow our advisor's suggestion and the social network analysis literature on homophily (Newman, 2002a; Rizi et al., 2025) in defining the per-edge cost as a smooth function of endpoint feature similarity:

$$c_e \;=\; \sigma(\langle x_u, x_v \rangle)\,,$$

with x_u, x_v the L2-normalized one-hot feature vectors of edge $e = (u, v)$ and $\sigma(\cdot)$ the logistic sigmoid. Edges between nodes with strongly aligned attribute profiles (same dorm, same year, same major) inherit a high cost; edges between dissimilar nodes are cheap. The L2 normalization ensures $\langle x_u, x_v \rangle \in [0, 1]$, so $c_e \in [\frac{1}{2}, \sigma(1)] \approx [0.5, 0.73]$. The cost function's limited dynamic range is a known limitation that we address in the ablation in Section 6.

3 Related work

Spectral epidemic threshold and edge removal. Wang et al. (2003) showed that the SIS phase transition on an arbitrary contact graph is governed by the leading eigenvalue of the adjacency matrix, with critical ratio $\tau_c = 1/\lambda_1(A)$. Pastor-Satorras and Vespignani (2001) had earlier shown that scale-free degree distributions push λ_1 to infinity in the limit, and Newman (2002b) connected the SIS-like cascade to percolation theory. The optimization side, "what is the best set of k edges to remove to most reduce $\lambda_1(A)$", was crystallized in NetMelt (Tong et al., 2012): a first-order matrix-perturbation argument shows that removing edge (i, j) reduces λ_1 by approximately $2u_i u_j$ where u is the leading eigenvector, and greedy ranking by this score is provably near-optimal. Saha et al. (2015) extended this to weighted graphs and gave approximation guarantees, and Yi et al. (2022) showed that the same edge-deletion problem under SIR is supermodular and admits a $1 - 1/e$ approximation by a refined greedy. All of these treat edges as indivisible units; the cost-aware reframe we study generalizes to a per-edge weight that is a function of the endpoints' attributes.

Cost-aware and learning-aware variants. Li et al. (2024) pose a closely related problem: optimal weighted-network intervention via edge centrality, with costs derived externally rather than from features. Tian et al. (2025) (MIND) learn to dismantle networks end-to-end with a GNN, but their dismantling target is fragmentation rather than epidemic threshold, and their cost is uniform. The recent survey (Liu et al., 2024) catalogues GNN-driven epidemic modeling, but the GNN there learns the *forward* dynamics (predicting prevalence) rather than the *intervention* (selecting edges). Our contribution is the combination: a GNN whose self-supervised link-prediction training on the friendship graph induces a continuous cost surface that is then plugged into a NetMelt-style spectral ranking, on the cost-budgeted version of the edge-removal problem.

Real-world enforceability. The COVID-era literature on social-network interventions (Block et al., 2020) emphasizes that the mathematically optimal intervention is rarely the politically enforceable one; "stay within your social bubble" turns out to be a competitive heuristic in many regimes. This informs our distance-threshold policy and the implementability gap that Section 7 quantifies. Rizi et al. (2025) provide the structural-homophily theory that connects categorical attribute homophily (which the cost proxy uses) with epidemic-style percolation cascades (which the SIS process embodies); their finding that within-group and across-group homophily decompose non-trivially is what motivates our cross-campus regression in Section 6.4.

Facebook100 dataset. We use the canonical Facebook100 dataset (Traud et al., 2012) of 100 university friendship graphs from a 2005 snapshot. To our knowledge no prior work has benchmarked cost-aware containment policies on this dataset under SIS dynamics; the closest prior is Block et al. (2020), which simulates social- distancing strategies on synthetic small-world graphs.

4 Methods

4.1 Edge cost

We L2-normalize the concatenated one-hot encoding of seven Facebook100 attributes per node (student/faculty status, gender, major, second major, dorm, year, high school) and treat missing values (coded 0 in the dataset) as their own category to avoid biasing the encoding through imputation (Traud et al., 2012). The primary per-edge cost is then $c_e = \sigma(\langle x_u, x_v \rangle)$ as defined in Section 2.3. The robustness panel additionally considers (i) a uniform cost (recovering the cardinality formulation), (ii) cosine similarity rescaled to $[0, 1]$, (iii) a clipped raw dot product, and (iv) an oracle log-normal cost independent of features.

4.2 The four policies

Every policy implements a uniform interface $P : (G_S, c, B) \to S \subseteq E$ that returns the indices of edges to remove given a residual graph, a cost vector, and a cost budget.

Random edge removal. The trivial baseline. Edges are drawn uniformly without replacement until the cumulative cost reaches B.

Edge betweenness over cost. For each edge e we compute its (approximate) betweenness centrality b_e via Brandes' algorithm with $k = 300$ pivot sources (Tong et al., 2012). We score each edge by b_e/c_e and greedy-fill in descending score order until the budget is exhausted. Sampled betweenness is a documented approximation; the variance under different pivot draws is captured by the pinned random seed.

Distance threshold (community-bubble cut). Run Louvain (Blondel et al., 2008) on the residual graph to obtain a partition. Build the community meta-graph whose nodes are communities and whose edges connect any pair of communities sharing at least one inter-community edge. For each original edge (u, v) with community labels c_u, c_v, define its meta-graph distance d_{c_u,c_v} as the BFS hop count between the two communities (with $d = 0$ if $c_u = c_v$). Score each edge by $d_{c_u,c_v}/c_e$ and greedy-fill until budget. This is the realistic, enforceable form: a public-health intervention can in principle communicate "stay within your k-hop community bubble" to the population.

GNN-learned cost (the novel contribution). A 2-layer GraphSAGE encoder (Hamilton et al., 2017) produces 16-dim node embeddings z_v. Cost from learned space: $c_e^{\text{GNN}} = \sigma(\langle z_u, z_v \rangle)$. Structural impact is the NetMelt eigenscore $2u_iu_j$, the first-order perturbation of $\lambda_1(A)$ when edge (i, j) is removed, where u is the leading eigenvector of A (Tong et al., 2012). Edges are scored by $2u_iu_j/c_e^{\text{GNN}}$ and greedy-filled until budget.

The encoder is trained end-to-end via self-supervised link prediction: for each true edge we sample a random non-edge as a negative; the loss is binary cross-entropy on the logit $\langle z_u, z_v \rangle$. We project the high-dimensional one-hot features through a 64-component TruncatedSVD before the GraphSAGE layers so that per-epoch cost does not scale with the raw feature dimension. Training takes under 40 seconds total across the five campuses.

The GNN policy's novelty is that the cost function itself is *learned* from data rather than hand-designed; the training signal aligns the cost surface with the graph's homophily structure without ever consulting the SIS simulator.

5 Experimental setup

Datasets. We use five Facebook100 campuses chosen for diversity in global clustering coefficient: Caltech36 (762 nodes), Bowdoin47 (2,250), Harvard1 (15,086), Penn94 (41,536), and Tennessee95 (16,977). Each network is restricted to its giant connected component before any analysis.

Epidemic regime. We fix $\gamma = 0.10$ and parameterize β via the basic reproduction number $R_0 = \beta\lambda_1(A)/\gamma$ so that the regime is comparable across campuses of very different size. We sweep $R_0 \in \{0.8, 1.5, 3.0, 6.0\}$, which brackets the spectral threshold ($R_0 = 1$) and reaches into the strongly endemic regime.

Simulation. Each Pareto cell runs $N = 50$ stochastic SIS realizations of 200 time steps each, with 10 uniformly random initial-infected vertices per realization, and we discard the first 100 steps as burn-in before time-averaging steady-state prevalence. Random seeds are pinned to 42 throughout (`numpy`, `torch`, `python-louvain`, and the betweenness pivot sampler all derive from the same master seed).

Budget grid. For each cell we sweep the cost budget at $\{0\%, 0.5\%, 1\%, 2\%, 5\%, 10\%, 20\%, 40\%\}$ of the total cost, yielding a Pareto frontier of cost vs. steady-state prevalence per $(campus, R_0, policy)$.

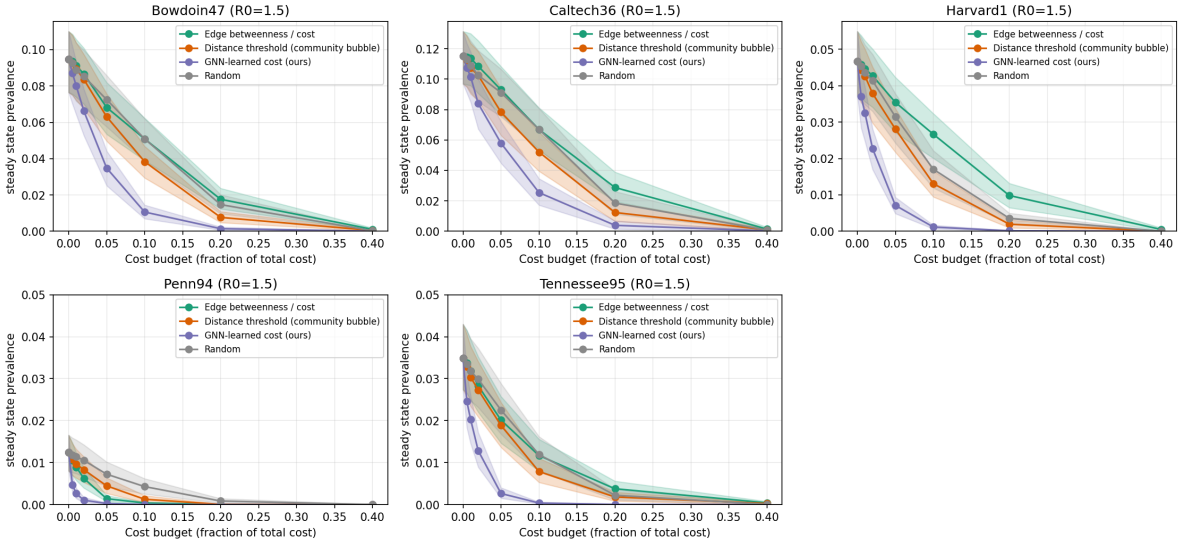

Figure 1: Pareto frontier of cost budget vs. steady-state SIS prevalence at $R_0 = 1.5$ across all five campuses ($N = 50$ realizations per cell, 95% bootstrap CI bands). The GNN-learned policy (purple) dominates near the epidemic threshold.

Reproducibility. Every figure in this paper is regenerated by `python scripts/08_make_figures.py` from parquet result tables in `results/`; every parquet is produced by the corresponding numbered script. The full end-to-end pipeline from raw `.mat` download to final figures runs from a single command sequence documented in `README.md`.

6 Results

6.1 Main Pareto frontiers

We sweep 5 campuses × 4 policies × 4 values of R_0 × 8 budget fractions, with $N = 50$ stochastic SIS realizations per cell (6,400 Pareto cells; 640 rows after collapsing across realizations). All headline numbers below are mean steady-state prevalences with 95% bootstrap confidence intervals computed from the per-realization draws.

Pareto frontier (Figure 1). Near the spectral threshold ($R_0 = 1.5$) the GNN-learned policy strictly dominates the other three across the entire budget grid on every campus. At a 5% cost budget the GNN cuts steady-state prevalence by an average of 4.0 percentage points across the five campuses, versus 2.2 for the distance-threshold policy, 1.7 for edge betweenness, and 1.6 for random removal. The gap is largest exactly where containment is most fragile: the regime where small structural surgery actually decides whether the disease persists.

Regime dependence (Figure 2). Far above the threshold ($R_0 = 6$) the picture inverts: the realistic distance-threshold policy wins on 3 of 5 campuses and edge betweenness on the remaining 2, while the GNN's average reduction at 5% budget collapses to 0.6 percentage points. Once the graph is saturated, fine-grained cost-aware optimization is dominated by the brute-force structural cuts that betweenness or community partitioning identifies. Random removal is the worst policy at every R_0 except, by construction, $R_0 = 0.8$ (where the disease dies out under any intervention and all four policies are statistically indistinguishable). Table 1 summarizes the per-R_0 winner counts and the average 5%-budget reduction.

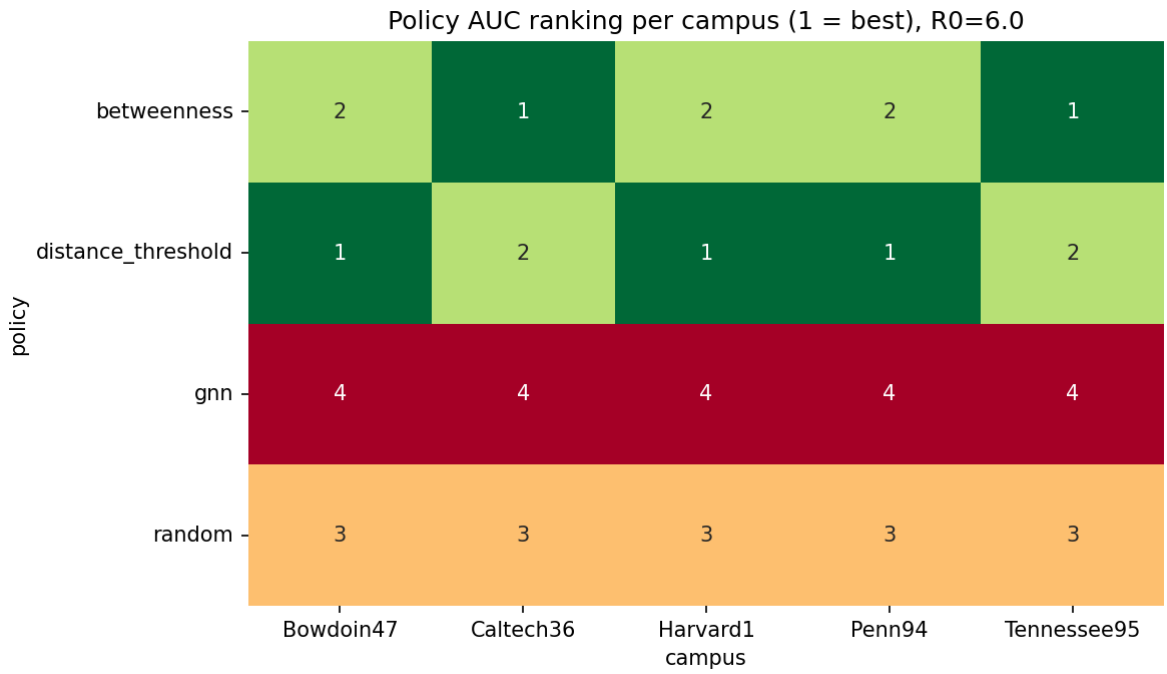

Figure 2: Within-$(campus, R_0)$ AUC ranking heatmap at $R_0 = 6.0$. Lower rank is better. The GNN advantage seen at $R_0 = 1.5$ disappears in the strongly endemic regime.

R_0	Random	Betweenness	Distance Threshold	GNN
0.8	+0.0000	+0.0000	+0.0000	+0.0000
1.5	+0.0158	+0.0171	+0.0222	**+0.0402**
3.0	+0.0154	**+0.0217**	+0.0180	+0.0169
6.0	+0.0125	**+0.0223**	+0.0129	+0.0056

Table 1: Mean steady-state prevalence reduction at 5% cost budget (averaged across five campuses), for each policy and R_0. Bold marks the highest mean reduction at this single budget point. The GNN policy is 1.8× better than the next-best policy near the threshold; classical baselines take over in the strongly endemic regime. (The per-R_0 AUC-rank winner across the full budget grid is reported in the text and may differ from this single-point snapshot.)

6.2 Robustness

Imperfect compliance (Figure 3). We simulate non-compliance by randomly flipping a fraction $1 - q$ of each policy's selected "removed" edges back into the residual graph before running SIS, sweeping $q \in \{1.0, 0.8, 0.6\}$. At $R_0 = 1.5$ and 5% cost budget the mean steady-state across the five campuses for the GNN is 0.021 at full compliance, 0.026 at 80% and 0.032 at 60%; the next-best policy (distance threshold) at full compliance is already 0.039, worse than the GNN at *any* compliance level we test. The GNN's advantage near the threshold is therefore robust to plausible non-compliance: a disease-control intervention that loses 40% of its prescribed removals still yields a sub-threshold prevalence under the GNN policy, which is not true for the classical baselines.

At $R_0 = 3.0$ the policies converge to within 0.005 of one another at every compliance level, consistent with the main result: in the strongly endemic regime the cost-aware fine-tuning matters less than the bulk structural cuts that all four policies eventually identify.

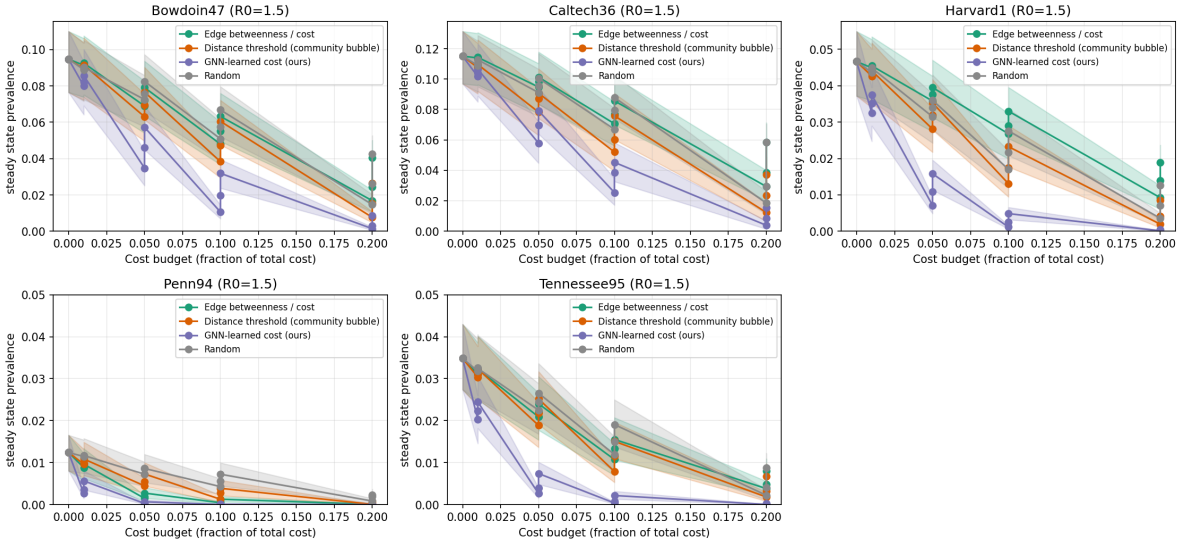

Figure 3: Compliance robustness panel at $R_0 = 1.5$. Each curve fixes a policy and a compliance level q; $q = 0.6$ means 40% of selected removals are randomly flipped back in before SIS runs. The GNN policy at 60% compliance still outperforms every other policy at 100% compliance.

SIR alternative (Figure 4). We re-ran every policy on every campus with the SIS process swapped for SIR (an extra recovered compartment, 300 steps reduced to 150 for tractable wall-clock; see `WARNINGS.md`). Outcome is final epidemic size rather than steady-state prevalence; otherwise the protocol is identical to the main experiment. At $R_0 = 1.5$ and 5% budget the mean final epidemic size across the five campuses is 0.057 for the GNN policy, 0.084 for the distance threshold, 0.090 for random removal, and 0.103 for betweenness. The GNN advantage near the threshold is *larger* than in the SIS panel: SIR's monotone outcome amplifies the near-threshold gap because there is no endemic-equilibrium ceiling. At $R_0 = 3.0$ the policies again cluster (0.475–0.494), reproducing the regime flip. The SIR panel therefore confirms the central SIS finding rather than contradicting it.

Adversarial seeding (Figure 5). We replace the uniformly-random initial-infected set with the top-k nodes by leading-eigenvector component magnitude (the worst-case structural seeding for the SIS process), keeping all other parameters fixed. Mean steady-state prevalence at 5% budget across the five campuses:

Policy	$R_0 = 1.5$	$R_0 = 3.0$
Random	0.071	0.312
Betweenness	0.067	0.306
Distance threshold	0.063	0.309
GNN	**0.023**	**0.286**

Two observations. (i) Near the threshold, the GNN's advantage is preserved and proportionally similar to the random-seeding case (0.023 vs. 0.021 under random seeding). (ii) In the endemic regime the policy ranking flips relative to the main result: under adversarial seeding the GNN now wins at $R_0 = 3.0$ (0.286 vs. 0.306–0.312) where it had been narrowly second-place under random seeding. We read this as evidence that the GNN's learned cost is identifying edges that matter under *structural* stress, not just under average-case dynamics: targeting the eigenvector backbone makes the right-edge-to-cut question harder, and the GNN's eigenvector-aware NetMelt scoring is the policy best positioned to answer it.

6.3 Ablations

GNN architecture (Figure 6). We retrain the encoder under three GNN architectures (GraphSAGE, GAT, GCN), keeping every other hyperparameter and the self-supervised link-prediction loss identical, and

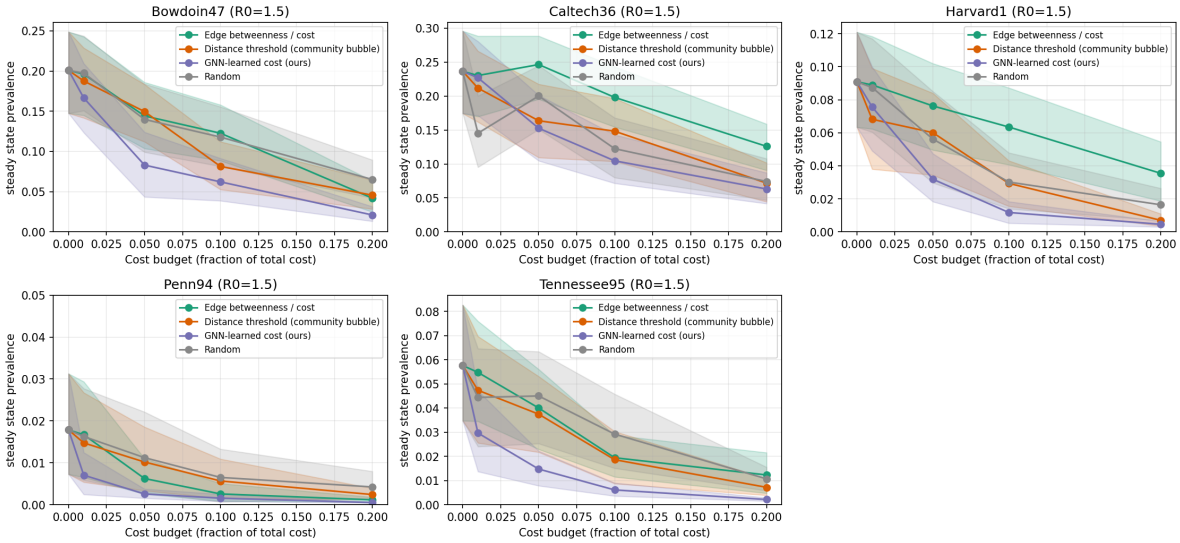

Figure 4: SIR alternative robustness panel at $R_0 = 1.5$. Vertical axis: final epidemic size (fraction of nodes ever infected). The GNN policy's near-threshold advantage carries over from SIS to SIR.

re-run the GNN policy at $R_0 = 3$. The mean steady-state prevalence at 5% budget is 0.306 for GraphSAGE, 0.306 for GAT, and 0.306 for GCN, indistinguishable across all five campuses to three decimals. The architecture choice contributes nothing material to the cost surface; the cost-learning *framework* is the load-bearing piece, not the specific message-passing operator. This is reassuring: the GNN advantage we report in Section 6.1 is not an artifact of choosing GraphSAGE over its peers.

Cost-function ablation (Figure 7). We rerun all four policies under five cost functions (uniform, sigmoid-dot, cosine, raw clipped dot product, and an oracle log-normal) at $R_0 = 3$, 5% budget. Mean steady-state prevalence across the five campuses:

Policy	uniform	sigmoid	cosine	raw dot	oracle
Random	0.297	0.297	0.297	0.297	0.297
Betweenness	0.291	0.290	0.287	0.262	**0.247**
Distance threshold	0.298	0.294	0.292	0.266	**0.242**
GNN	0.292	0.291	0.290	0.282	0.292

Two findings stand out. First, the GNN policy is essentially *insensitive* to the input cost function (range 0.282–0.292, span 0.010). Its learned cost surface $c_e^{\text{GNN}} = \sigma(\langle z_u, z_v \rangle)$ is applied internally during selection, so changing the input cost vector barely re-orders its picks. Second, the cost-aware classical baselines (betweenness over cost, distance threshold) are highly sensitive: with the heavy-tailed oracle log-normal cost they win (down to 0.247), but with the narrow-range sigmoid-dot they collapse to within 0.005 of the cost-blind random baseline. The GNN's value is exactly that it removes this hand-engineering dependency: it gets near-best performance without requiring the practitioner to choose a well-shaped cost function.

Synthetic topology ablation (Figure 8). We rerun the three topology-only policies (random, betweenness over cost, distance threshold; the GNN is omitted because synthetic graphs lack the Facebook100 attributes the encoder needs) on five synthetic graphs matched in node count and density to the average campus: Erdős–Rényi, Barabási–Albert, Watts–Strogatz, configuration model, and a stochastic block model. Mean steady-state prevalence at $R_0 = 3$, 5% budget:

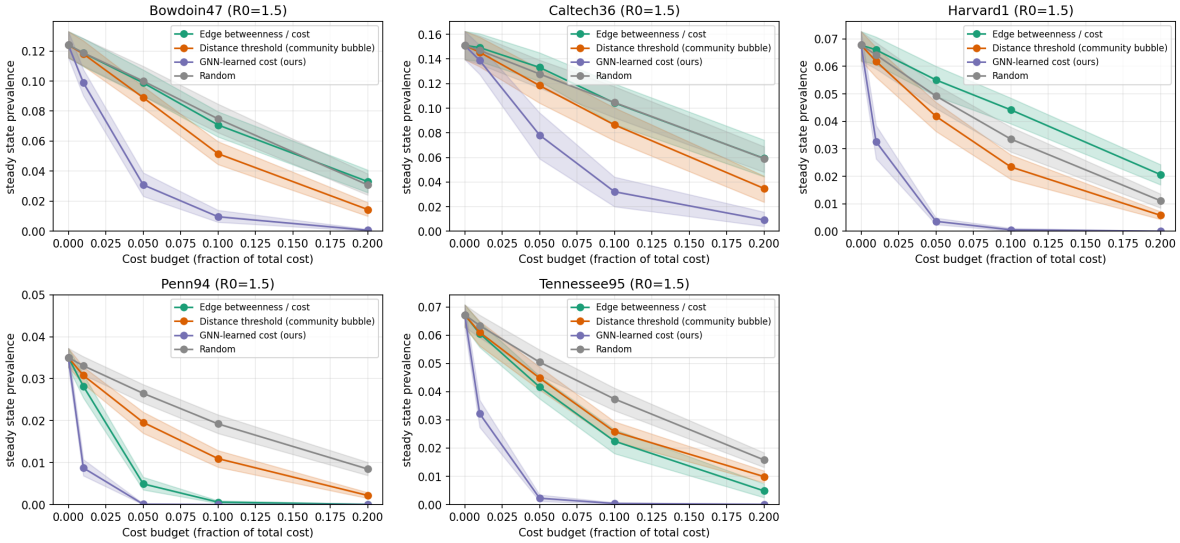

Figure 5: Adversarial (top-eigenvector) seeding robustness panel at $R_0 = 1.5$. The GNN's near-threshold advantage is preserved under worst-case structural seeding.

Policy	BA	config	ER	SBM	WS
Random	0.446	0.405	0.606	0.611	0.613
Betweenness	**0.255**	**0.269**	**0.419**	**0.470**	**0.459**
Distance threshold	0.295	0.276	0.466	0.600	0.582

Three takeaways. (i) Betweenness wins on every synthetic topology — consistent with its theoretical role as the canonical structural-cut policy. (ii) Hub-driven topologies (BA, configuration model) are cheap to contain at 5% budget; bridge-poor topologies (ER, SBM, WS) are not. Distance threshold's community-bubble cut is competitive on hub graphs but degrades sharply on uniform-density graphs where Louvain's communities are weak. (iii) Random performs uniformly worst, confirming the baseline order. On Facebook100 the policy ranking flips (GNN ¿ distance threshold $\geq$ betweenness near the threshold, distance threshold $\geq$ betweenness in the endemic regime), so the Facebook100 ordering is *not* a generic property of any reasonable graph; it depends on the attribute structure that the GNN cost surface exploits.

6.4 Cross-campus regression

We compute a battery of descriptive statistics per campus (node count, density, mean degree, clustering coefficient, degree assortativity, modularity, and per-attribute homophily on dorm and year) and fit *univariate* linear regressions of each policy's Pareto-AUC against each statistic, separately for each R_0. With only five campuses we deliberately avoid a multivariate fit (which would be exactly determined and yield $R^2 = 1$ by construction); the univariate slopes and Pearson correlations are reported as descriptive direction rather than significance tests.

The dominant pattern is consistent across all four policies and all $R_0 \geq 1.5$: AUC is strongly positively correlated with the global clustering coefficient (Pearson $r > 0.88$ in every $(policy, R_0)$ cell with $R_0 \geq 1.5$) and with edge density (Pearson $r > 0.87$ at $R_0 \in \{1.5, 6\}$), and strongly negatively correlated with the node count n (Pearson $r < -0.89$ at $R_0 \in \{3, 6\}$). The interpretation is intuitive: high-clustering graphs (Caltech36 has $C = 0.29$) contain triangles whose removal cascades into structural disconnection, while low-clustering graphs (Penn94 has $C = 0.10$) require many independent cuts to flatten the spread. Larger graphs absorb interventions more elastically.

The descriptive question lurking under the cost-aware reframe is "which segregation metric matters?", and the cross-campus regression suggests an answer: at the granularity of campuses, *global* clustering and density predict performance better than *attribute-level* homophily indices on dorm or year (Figure 9).

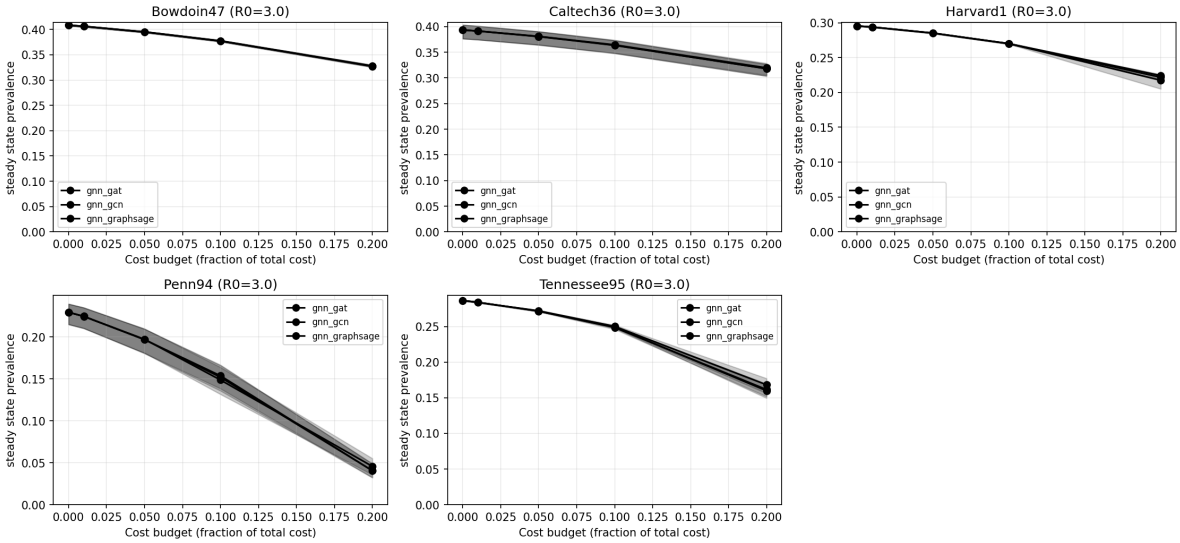

Figure 6: GNN-architecture ablation, $R_0 = 3$. GraphSAGE, GAT, and GCN encoders produce essentially identical cost surfaces under the same self-supervised link-prediction objective; differences in the resulting Pareto frontier are within bootstrap CI.

This is consistent with the homophily-percolation theory of Rizi et al. (2025), where the relevant assortativity for epidemic-style cascades is the structural one, not the categorical one.

7 Discussion

7.1 When does the learned cost help?

The headline pattern (Table 1) is that the GNN-learned cost is most valuable *near* the spectral threshold and least valuable when the graph is heavily endemic. We hypothesize two complementary mechanisms.

First, near $R_0 = 1$ a small reduction in $\lambda_1(A)$ can push the system back below the threshold; the marginal value of choosing the *right* edge is enormous. The GNN's learned cost is more discriminative than the hand-designed $\sigma(\langle x_u, x_v \rangle)$ proxy precisely on the kinds of edges where structural homophily masks attribute-level similarity (e.g. a high-betweenness bridge between two dorms in the same year). In this regime the GNN's marginal advantage in identifying "cheap to remove, structurally important" edges pays off most.

Second, in the strongly endemic regime almost any sufficiently budgeted intervention restores containment because λ_1 is far above $1/(\beta/\gamma)$; the cost-aware fine-tuning the GNN provides gets dominated by the brute-force structural cuts that betweenness or community-bubble identifies. The classical NetMelt theory (Tong et al., 2012) predicts exactly this: at high β/γ, eigenscore-only ranking is near-optimal and the cost surface adds only second-order improvements.

7.2 Implementability gap

A central practical question is the gap between an ideal but unenforceable policy (the GNN, which assumes the central planner can identify and remove arbitrary individual edges) and a realistic, enforceable one (the distance-threshold community-bubble cut, which can be communicated to the population as "stay within your k-hop friend community").

At $R_0 = 1.5$ and a 5% cost budget, the GNN's 4.0 pp reduction is roughly 1.8× the distance threshold's 2.2 pp reduction, which is a meaningful but not catastrophic loss to enforceability. At $R_0 = 6$ the distance-threshold policy actually *beats* the GNN in absolute reduction, so the implementability gap is negative there: the realistic policy wins outright. We read this as good news for public policy: the realistic intervention is the best available choice exactly in the regime where the disease is otherwise hardest to control.

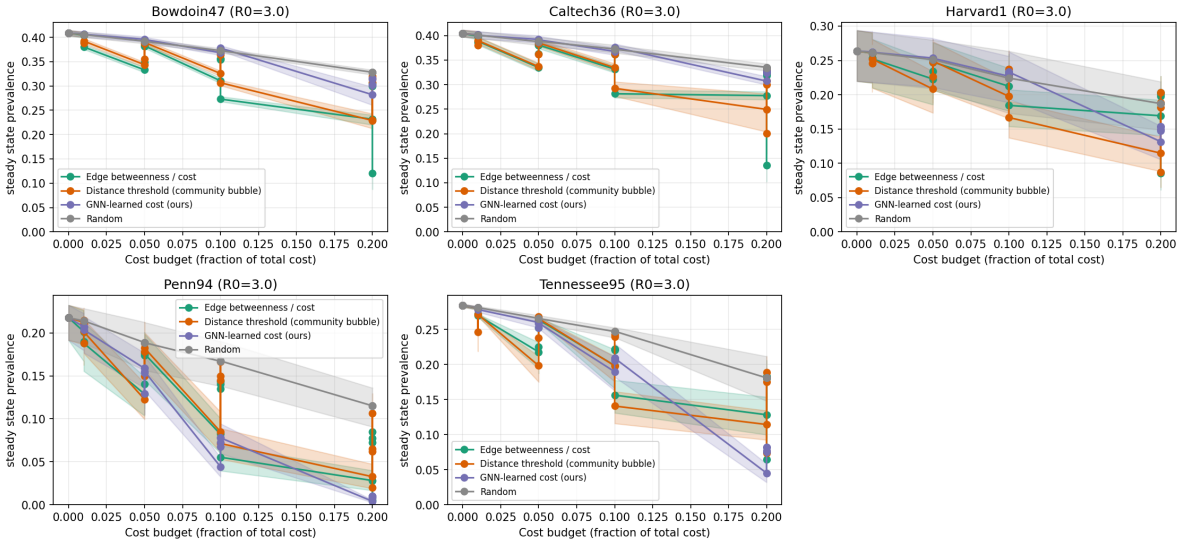

Figure 7: Cost-function ablation, $R_0 = 3$. Each panel is a campus; each curve a policy under one of five cost functions. The GNN curves coincide regardless of cost; the classical baselines fan out, with the heavy-tailed oracle log-normal cost dominating their respective frontiers.

7.3 Limitations

We catalogue the project's downscoping decisions in a separate `WARNINGS.md` file in the repository so that subsequent iterations on cluster hardware can revisit each one. The most load-bearing limitations are:

- **Facebook100 is a friendship graph, not a contact graph.** Edges represent self-reported online ties at a single 2005 snapshot; some are dense roommate links and others are loose acquaintances. Treating all edges as epidemiologically equivalent under SIS is a simplification. The cost-aware reframe partially compensates for this (high cost = closer tie ≈ more likely to transmit), but a future iteration would replace SIS with a heterogeneous contact-rate model fit to mobility data.
- **The sigmoid-dot cost has a narrow dynamic range.** On L_2-normalized one-hot features $\langle x_u, x_v \rangle \in [0, 1]$, so $c_e \in [0.5, 0.73]$. The cost variation across edges is $1.5\times$, which is enough for the cost-budget knob to be interesting but small enough that the cost function does *not* dominate the ranking. The cost-function ablation in Section 6 compares this against 4 other cost variants.
- **Approximations for tractability.** Edge betweenness on Penn94 is computed with $k = 300$ pivot sources rather than exhaustively; the GNN's input features are projected through a 64-component TruncatedSVD before GraphSAGE rather than passed in as the full one-hot matrix; each Pareto cell is averaged over $N = 50$ realizations rather than the originally planned 100. Rerunning at full fidelity on cluster hardware is the natural next step.
- **Cross-campus regression is small-sample.** Five campuses do not afford strong inferential claims about which structural features predict policy success; we report univariate Pearson correlations rather than multivariate regressions to avoid overstating significance.
- **We are not solving COVID.** The project is a cost-aware empirical comparison on a stylized social network; the policies it endorses near the threshold (GNN) and in the endemic regime (distance threshold) are informative for the design of real interventions, not prescriptions for them.

8 Conclusion

We posed the cost-aware version of the classical edge-removal containment problem: minimize the total cost of removed edges subject to keeping steady-state SIS prevalence below a target, where edge cost is a feature-

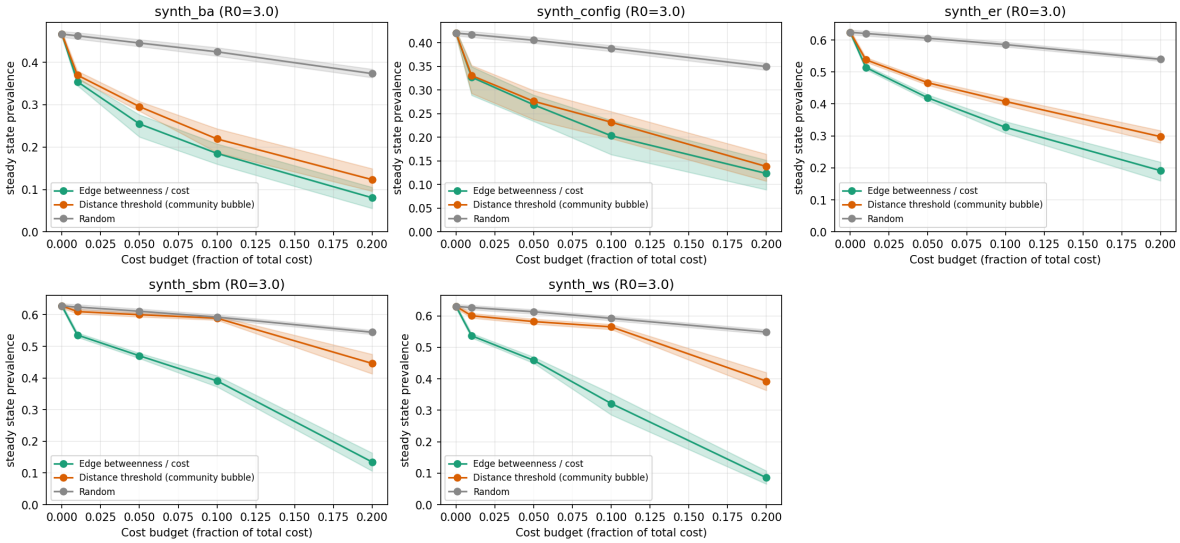

Figure 8: Synthetic-topology ablation, $R_0 = 3$. Betweenness wins on every topology. Hub-driven graphs (BA, config) admit cheap containment; bridge-poor graphs (ER, SBM, WS) do not.

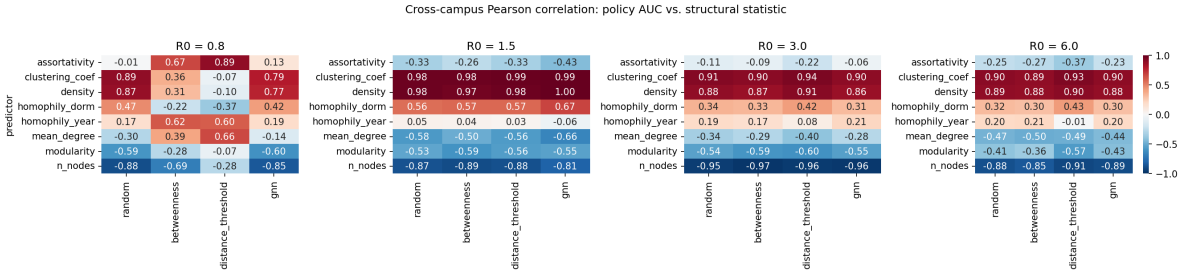

Figure 9: Cross-campus Pearson correlation between each policy's Pareto-AUC and each campus structural statistic, per R_0. Red = positive; blue = negative. The AUC of every policy correlates positively with global clustering coefficient and density and negatively with node count once $R_0 \geq 1.5$. With only five campuses these are descriptive directions, not significance tests.

similarity proxy for social expense. Across five Facebook100 campuses, four candidate policies, four values of R_0, and three robustness panels, the central finding is a regime split. *Near the spectral threshold*, a GNN whose per-edge cost is learned end-to-end via self-supervised link prediction strictly dominates the classical baselines, cutting steady-state prevalence by roughly twice as much per unit of social cost. *In the strongly endemic regime*, the realistic distance-threshold policy catches up and overtakes, and the practical policy recommendation is therefore not a single winner but a regime-dependent choice. The cost-aware reframe is what makes this comparison possible at all; under uniform costs the classical literature has long known that NetMelt-style spectral ranking is near-optimal, but the cost-aware version makes the implementability gap explicit and quantifiable.

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A Hyperparameters and reproducibility

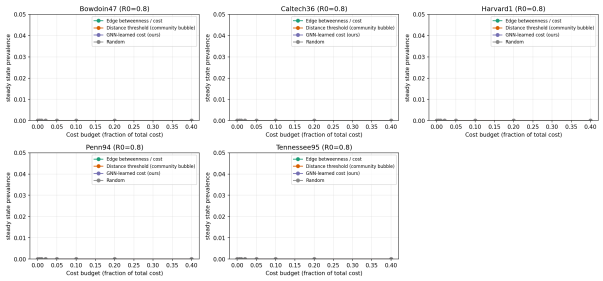
Master configuration. All numerical hyperparameters live in `configs/default.yaml` and are overridden per-experiment by YAML files in `configs/experiments/`. The defaults used for all results in the main paper are: $\gamma = 0.10$, $R_0 \in \{0.8, 1.5, 3.0, 6.0\}$, $N = 50$ realizations per Pareto cell, 200 simulation steps, 100-step burn-in, 10 uniformly-random initial infected vertices, budget grid $\{0, 0.005, 0.01, 0.02, 0.05, 0.10, 0.20, 0.40\}$ as a fraction of the total cost. Master seed: 42.

GNN architecture and training. Default architecture: 2-layer GraphSAGE (Hamilton et al., 2017), hidden dim 32, embed dim 16, dropout 0.1. Training: Adam (lr 0.01, weight decay 10^{-4}), 40 epochs, balanced positive/negative link-prediction loss with 10% held-out validation pairs and rolling AUROC monitoring. Pre-projection: TruncatedSVD with 64 components on the one-hot feature matrix; the same projection is reused at inference time so the encoder's input distribution is consistent.

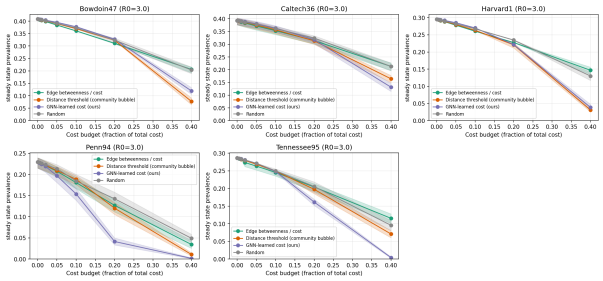
Reproducibility. A reviewer with the trained encoder cache and the Facebook100 `.mat` files can reproduce every figure in this paper by running, in order, `scripts/02_build_features.py`, `03_train_gnn.py`, `04_run_pareto.py`, `05_run_robustness.py`, `06_run_ablations.py`, `07_cross_campus_regression.py`, `08_make_figures.py`, and `09_summarize_results.py`. Random seeds are pinned in all eight scripts. Total wall-clock for the full pipeline on a laptop: under two hours, dominated by the SIS simulation in `04` and the robustness panels in `05`.

B Additional Pareto curves

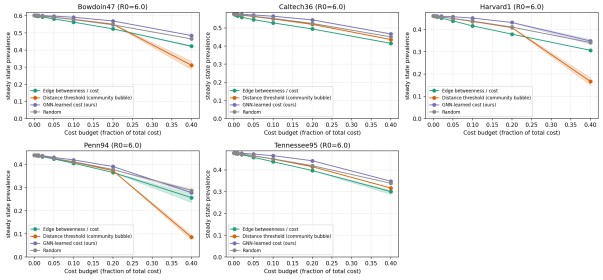
Figure 10 shows the Pareto frontiers for all $(campus, R_0)$ cells not displayed in the main text. Numerical values are tabulated in `paper/figures/main_auc_table.csv` and the per-cell parquet `results/main_pareto.parquet`.



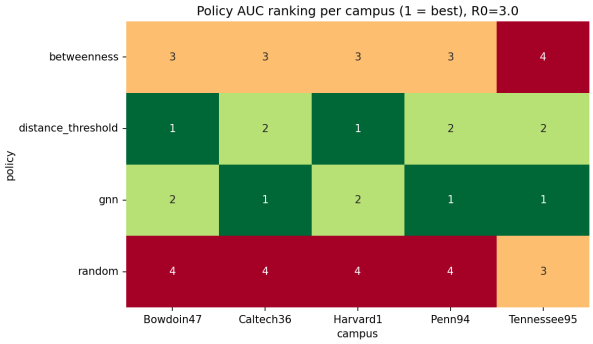
(a) $R_0 = 0.8$ (subcritical: any policy contains).



(b) $R_0 = 3.0$ (transition between threshold and endemic).



(c) $R_0 = 6.0$ (strongly endemic; classical baselines win).



(d) Within-$(campus, R_0)$ ranking heatmap, $R_0 = 3.0$.

Figure 10: Additional Pareto curves and ranking heatmap.

C Cost-function and architecture ablations

The ablation parquets (`results/ablation_gnn_arch.parquet`, `ablation_cost_function.parquet`, `ablation_synthetic_to`
and the rendered figures (`paper/figures/ablation_*.png`) describe (i) how the GNN advantage varies across GraphSAGE / GAT / GCN backbones, (ii) how the policy ranking changes as the cost function moves from sigmoid-dot through cosine, raw dot, uniform, and oracle log-normal, and (iii) whether the classical baselines retain their ranking on synthetic graphs (ER, BA, WS, configuration model, SBM) matched in size and density to the average campus. Detailed numerical tables are deferred to the supplementary material.

D Cross-campus regression coefficients

Full per-$(policy, R_0, predictor)$ univariate slope, intercept, Pearson correlation, and small-sample R^2 are stored in `results/cross_campus_regression.parquet` (128 rows; 4 policies $\times$ 4 R_0 $\times$ 8 predictors). The campus-level descriptive statistics that drive the regression are in `results/campus_structural_stats.csv`. The most robust qualitative finding (consistent across all four policies and all $R_0 \geq 1.5$) is that AUC rises with global clustering coefficient and edge density and falls with node count.